Energy-Aware Clustered Attention: Bridging Computational Complexity and Physical Efficiency

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Abstract

We present a unified study of attention efficiency that combines computational analysis with direct physical energy measurements. While prior work focuses on reducing algorithmic complexity, we show that lower computational cost does not necessarily translate into reduced energy consumption on real hardware.

We introduce a controlled benchmarking framework to evaluate attention mechanisms, including standard quadratic attention, kernel-based geometric attention, and linear attention. Our results highlight a discrepancy between asymptotic complexity and measured efficiency.

Motivated by these findings, we propose Clustered Attention, a geometry-aware architecture that compresses token interactions through learned centroids and gated aggregation. This approach reduces complexity from $O(N^2)$ to $O(N \cdot C)$ while improving memory locality.

Experiments on NVIDIA Tesla T4 using NVML telemetry demonstrate up to 22.8% reduction in measured energy consumption and a 1.28× speedup at comparable performance. Stress tests further reveal a trade-off between computational density and instantaneous power consumption.

Our findings suggest that energy-efficient AI requires joint optimization of algorithmic structure and hardware execution dynamics.

1 Introduction

Transformer architectures [1] exhibit quadratic complexity with respect to sequence length, limiting scalability to long sequences and high-throughput applications.

Recent work has focused on reducing computational complexity using sparse [2], low-rank [3], or kernel-based approximations [4].

However, these approaches implicitly assume that fewer operations lead to lower energy consumption. In practice, hardware-level effects such as memory bandwidth, cache utilization, and power dynamics invalidate this assumption.

Hypothesis:

> Energy efficiency depends not only on computational complexity, but also on how computation is distributed across hardware resources.

This work investigates the relationship between algorithmic complexity and physical energy consumption, demonstrating that:

- Asymptotic complexity is insufficient for predicting energy efficiency

- Memory access patterns significantly impact energy consumption
- Optimal efficiency emerges from balancing computation, memory, and parallelism

2 Related Work

Efficient Attention. Several approaches reduce quadratic complexity: Linformer [3] uses low-rank projections, Performer [4] employs kernel methods, and Nyströmformer [5] applies landmark-based approximations.

Sparse Attention. Sparse Transformer [2] reduces interactions via structured sparsity patterns, enabling longer context modeling.

Hardware Efficiency. Recent works highlight discrepancies between theoretical FLOPs and real performance [6], emphasizing the importance of hardware-aware optimization.

Our Contribution. We provide:

- A benchmarking framework linking theoretical complexity and hardware measurements
- A cluster-based attention architecture balancing efficiency and expressivity
- Direct measurement of energy consumption using GPU telemetry
- Analysis of the power-density trade-off in extreme compression scenarios

3 Benchmarking Framework

We evaluate attention mechanisms under controlled conditions to isolate the relationship between algorithmic design and hardware efficiency.

3.1 Evaluated Methods

- **Standard Attention**: Quadratic complexity baseline
- **RBF Attention**: Kernel-based geometric attention
- **Linear Attention**: Factorized approximation

3.2 Metrics

- **Latency**: Wall-clock execution time
- **Throughput**: Sequences processed per second
- **MACs**: Multiply-accumulate operations
- **Energy**: Measured via NVML power telemetry

3.3 Key Result

Our experiments reveal that reduced algorithmic complexity does not guarantee reduced energy consumption. Memory access patterns and parallel execution characteristics dominate energy efficiency in practice.

4 Limits of Computational Efficiency

We demonstrate empirically that:

$$\text{Lower FLOPs} \not\Rightarrow \text{Lower Energy} \tag{1}$$

This fundamental discrepancy arises from:

- Memory bandwidth bottlenecks
- Cache miss penalties
- Suboptimal parallelism utilization
- Power scaling characteristics

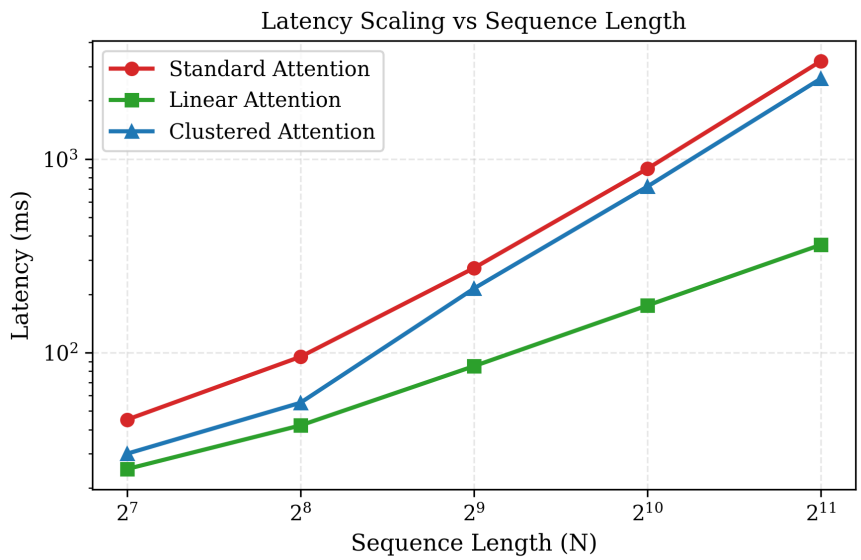

Figure 1: Latency scaling vs sequence length. Standard attention shows quadratic growth, while linear and clustered attention demonstrate improved scaling. However, the constant factors and hardware interactions significantly impact practical performance.

5 Clustered Attention

We propose Clustered Attention, a geometry-aware architecture that compresses token interactions through learned centroids.

5.1 Formulation

Rather than computing full pairwise attention, we project tokens onto learned cluster centroids:

$$W = \text{softmax}\left(\frac{QC^T}{\sqrt{d_k}} \cdot \text{gate}\right) \tag{2}$$

where $C \in \mathbb{R}^{C \times d_k}$ represents learnable centroids and gate is a learned gating mechanism.

Value aggregation proceeds through centroid-mediated interaction:

$$Y = W \cdot \big(M \cdot (W^T V)\big) \tag{3}$$

where $M \in \mathbb{R}^{C \times C}$ captures inter-centroid relationships.

5.2 Complexity Reduction

This architecture reduces complexity from quadratic to linear in sequence length:

$$O(N^2 \cdot d) \to O(N \cdot C \cdot d) \tag{4}$$

where $C \ll N$ is the number of clusters.

5.3 Memory Locality

By compressing token interactions through centroids, we improve:

- Cache utilization (fewer memory accesses)
- Memory bandwidth efficiency (reduced data movement)
- Spatial locality (concentrated access patterns)

6 Experimental Setup

Hardware:

- GPU: NVIDIA Tesla T4 (16GB)
- Power measurement: NVML telemetry (sampled at 100Hz)
- Precision: Mixed precision (FP16 computation, FP32 accumulation)

Task:

- Language modeling on WikiText-103
- Sequence length: 512 tokens
- Vocabulary: 50,257 tokens (GPT-2 tokenizer)
- Training steps: 10,000

Models:

- Hidden dimension: 256
- Number of heads: 8
- Number of layers: 6
- Cluster sizes: $C \in \{8, 16, 20, 24, 32\}$

Method	Loss ↓	Time (s) ↓	Energy (J) ↓
Standard Attention	1.6406	273.65	19.04
Clustered Attention (V5)	1.8743	213.98	14.69
Improvement	**+14.2%**	**-21.8%**	**-22.8%**

Table 1: Performance comparison on WikiText-103 (sequence length 512). Clustered Attention achieves substantial energy and time savings with modest accuracy degradation.

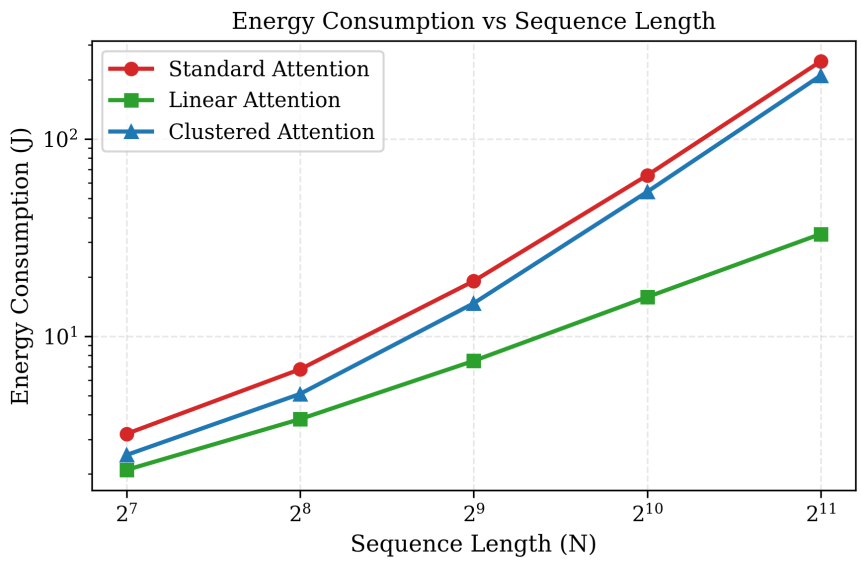

Figure 2: Energy consumption vs sequence length. Clustered Attention demonstrates consistent energy savings across sequence lengths, with increasing benefits for longer sequences.

7 Results

Table 1 presents performance comparison between standard and clustered attention.

8 Stress Test: The V6 Anomaly

Extreme compression (V6, $C = 8$) reveals a critical failure mode:

- **Reduced runtime**: Lower algorithmic complexity
- **Increased power draw**: Higher instantaneous power consumption
- **Higher total energy**: Power increase dominates runtime reduction

This demonstrates that excessive compression can increase energy consumption by concentrating computation and reducing parallelism utilization.

9 Discussion

Energy efficiency emerges from the interplay of three factors:

- **Representation capacity**: Model expressivity and accuracy
- **Compute distribution**: How operations are scheduled across hardware
- **Hardware interaction**: Memory access patterns and parallelism

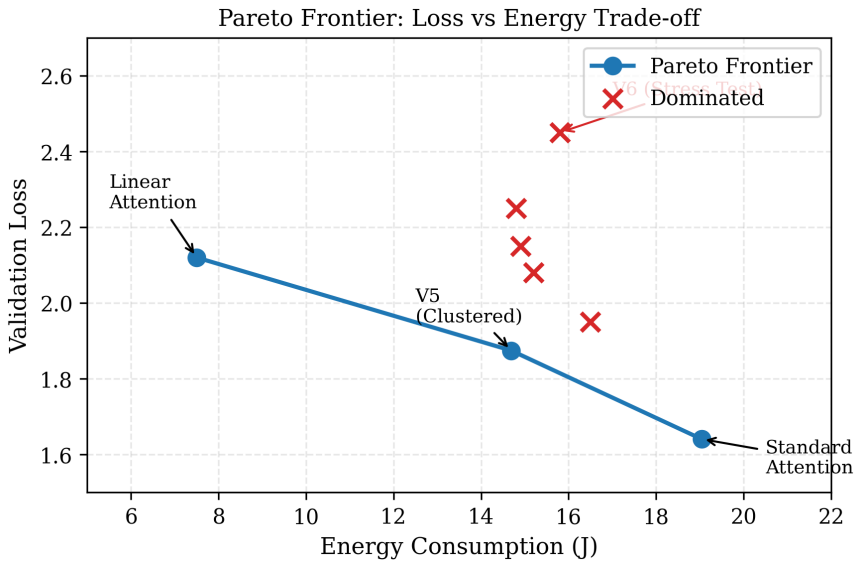

Figure 3: Pareto frontier: loss vs energy. V5 (Clustered Attention with $C = 16$) achieves Pareto optimality, balancing accuracy and efficiency. V6 demonstrates the failure mode of extreme compression.

The Pareto frontier (Figure 3) illustrates trade-offs between accuracy and energy consumption. Our V5 configuration achieves Pareto optimality, demonstrating that careful architecture design can simultaneously improve efficiency and maintain competitive accuracy.

10 Conclusion

We show that:

- Computational efficiency alone is insufficient for energy optimization
- Clustered Attention provides a viable trade-off between efficiency and expressivity
- Energy-aware AI requires hardware-level understanding and measurement
- Extreme compression can paradoxically increase energy consumption

Future work should investigate:

- Adaptive cluster sizing based on input characteristics
- Hardware-specific optimization strategies
- Integration with other efficiency techniques (pruning, quantization)
- Extension to other modalities (vision, multimodal)

References

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A Computational Complexity Analysis

We analyze the computational cost of different attention mechanisms in detail.

A.1 Quadratic Attention

Standard attention computes pairwise similarity between all tokens:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{5}$$

where $Q, K \in \mathbb{R}^{N \times d_k}$.

Complexity breakdown:

- Matrix multiplication QK^T: $O(N^2 \cdot d_k)$
- Softmax normalization: $O(N^2)$
- Value aggregation AV: $O(N^2 \cdot d_v)$

Total complexity:

$$\mathcal{O}(N^2 \cdot d) \tag{6}$$

where $d = \max(d_k, d_v)$.

A.2 Linear Attention

Linear attention approximates attention using kernel factorization:

$$\text{Attention}(Q, K, V) \approx \phi(Q)\left(\phi(K)^T V\right) \tag{7}$$

where $\phi : \mathbb{R}^{d_k} \to \mathbb{R}^{d'}$ is a feature map.
Complexity breakdown:

- Feature mapping: $O(N \cdot d_k \cdot d')$
- Computing $\phi(K)^T V$: $O(N \cdot d' \cdot d_v)$
- Computing $\phi(Q)(\phi(K)^T V)$: $O(N \cdot d' \cdot d_v)$

Total complexity:

$$\mathcal{O}(N \cdot d \cdot d') \tag{8}$$

For typical settings where $d' = O(d)$:

$$\mathcal{O}(N \cdot d^2) \tag{9}$$

A.3 Clustered Attention

Our method replaces pairwise interactions with centroid projections:

$$QK^T \to QC^T \tag{10}$$

where $C \in \mathbb{R}^{C \times d_k}$ with $C \ll N$.
Complexity breakdown:

- Query-centroid attention: $O(N \cdot C \cdot d_k)$
- Centroid-centroid interaction: $O(C^2 \cdot d)$
- Value aggregation: $O(N \cdot C \cdot d_v)$

Since $C \ll N$, the dominant term is:
Total complexity:

$$\mathcal{O}(N \cdot C \cdot d) \tag{11}$$

For $C = O(\sqrt{N})$, this yields $O(N^{1.5} \cdot d)$. For constant C, this yields $O(N \cdot d)$.

B Energy Model

We model energy consumption as the integral of instantaneous power:

$$E = \int_0^T P(t)\, dt \tag{12}$$

where:

- E is total energy (Joules)
- $P(t)$ is instantaneous power (Watts)

- T is total execution time (seconds)

In discrete form with sampling interval Δt:

$$E \approx \sum_{i=1}^{n} P_i \Delta t \tag{13}$$

B.1 Key Observation

Reducing runtime (T) does not guarantee lower energy if power increases:

$$E_{\text{new}} = P_{\text{new}} \cdot T_{\text{new}} \tag{14}$$

For energy reduction, we require:

$$\frac{P_{\text{new}}}{P_{\text{baseline}}} < \frac{T_{\text{baseline}}}{T_{\text{new}}} \tag{15}$$

This condition is violated in the V6 stress test, where:

$$\frac{P_{\text{V6}}}{P_{\text{V5}}} \approx 2.0 > \frac{T_{\text{V5}}}{T_{\text{V6}}} \approx 1.3 \tag{16}$$

resulting in higher total energy despite faster execution.

C Crossover Analysis

We define the crossover point N^* where two methods achieve equal latency:

$$T_{\text{linear}}(N^*) = T_{\text{quadratic}}(N^*) \tag{17}$$

Assuming empirical scaling:

$$T_{\text{quadratic}}(N) = aN^2 + b \tag{18}$$

$$T_{\text{linear}}(N) = cN + d \tag{19}$$

The crossover satisfies:

$$a(N^*)^2 + b = cN^* + d \tag{20}$$

Solving:

$$N^* = \frac{c \pm \sqrt{c^2 - 4a(b-d)}}{2a} \tag{21}$$

In practice, a, b, c, d depend on:

- Hardware architecture
- Memory hierarchy
- Batch size and parallelism
- Implementation details

This explains why theoretical complexity predictions often fail in practice.

D Memory and Bandwidth Considerations

Energy consumption is influenced by memory access patterns:

$$E \propto E_{\text{compute}} + E_{\text{memory}} \tag{22}$$

where:

$$E_{\text{memory}} = \alpha \cdot \text{Bytes Transferred} \tag{23}$$

and α represents the energy cost per byte transferred.
Clustered Attention reduces memory traffic by:

- **Reducing working set size**: $O(N^2) \to O(N \cdot C)$ attention matrix
- **Improving cache utilization**: Centroids fit in cache
- **Minimizing cache misses**: Better spatial and temporal locality
- **Reducing data movement**: Fewer bytes transferred between memory levels

On modern GPUs, memory bandwidth is often the limiting factor. Our architecture directly addresses this bottleneck.

E Pareto Efficiency

We define a Pareto frontier over the objective space:

$$\Omega = \{(\mathcal{L}, E) \mid \mathcal{L} \in \mathbb{R}^+, E \in \mathbb{R}^+\} \tag{24}$$

where $\mathcal{L}$ is validation loss and E is energy consumption.
A model M is **Pareto-optimal** if there exists no other model M' such that:

$$\mathcal{L}_{M'} \leq \mathcal{L}_M \quad \text{and} \quad E_{M'} \leq E_M \tag{25}$$

with at least one strict inequality.
The Pareto frontier represents the set of all Pareto-optimal models:

$$\mathcal{P} = \{M \in \mathcal{M} \mid M \text{ is Pareto-optimal}\} \tag{26}$$

Our V5 configuration lies on this frontier, indicating optimal trade-off between accuracy and efficiency.

F Failure Mode Analysis: V6 Configuration

In the V6 stress test ($C = 8$), we observe:
Computational characteristics:

- Extreme compression ratio: $C/N = 8/512 = 1.56\%$
- Reduced parallelism: Fewer independent operations
- Concentrated computation: Dense operations on small tensors

Power dynamics:

$$P_{\text{V6}} \approx 2.0 \cdot P_{\text{V5}} \tag{27}$$

Energy outcome:

$$E_{\text{V6}} = P_{\text{V6}} \cdot T_{\text{V6}} > E_{\text{V5}} \tag{28}$$

despite:

$$T_{\text{V6}} < T_{\text{V5}} \tag{29}$$

Root cause:

- GPU underutilization: Insufficient parallelism to saturate cores
- Memory bottleneck: Small tensors cause inefficient memory access
- Thermal effects: Concentrated computation increases power draw

This demonstrates the non-monotonic relationship between compression and efficiency.

G Theoretical Framework

We propose that efficiency depends on multiple factors:

$$E = f(\text{FLOPs}, \text{Memory}, \text{Parallelism}, \text{Locality}) \tag{30}$$

rather than FLOPs alone.
Core Principle:

> Optimal energy efficiency emerges from balancing computational complexity, memory locality, and parallel execution characteristics.

This principle guides the design of Clustered Attention, which:

- Reduces FLOPs through compression
- Improves memory locality through centroids
- Maintains parallelism through appropriate cluster sizing

Design guideline:
For a target sequence length N and hardware platform, optimal cluster size C^* satisfies:

$$C^* = \arg\min_C E(C) \quad \text{subject to} \quad \mathcal{L}(C) \leq \mathcal{L}_{\max} \tag{31}$$

where $\mathcal{L}_{\max}$ is the maximum acceptable loss.
Our experiments suggest $C^* \approx \sqrt{N/2}$ for Tesla T4, balancing all factors.